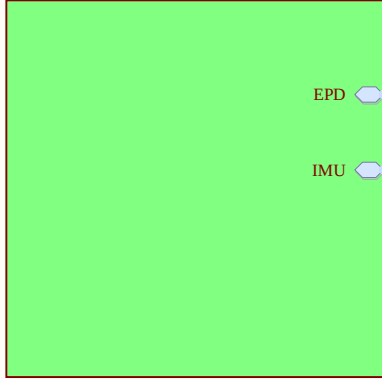
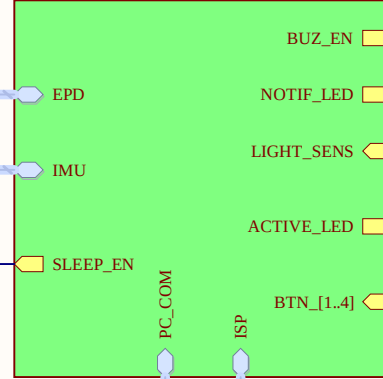


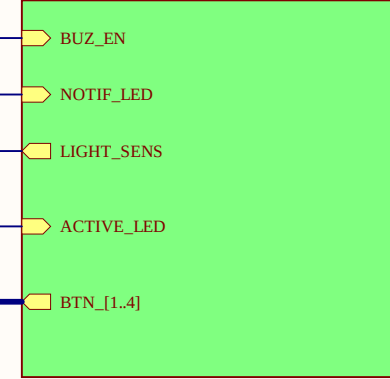
IMU
imu_and_epd.SchDoc



MCU
peep_avr.SchDoc



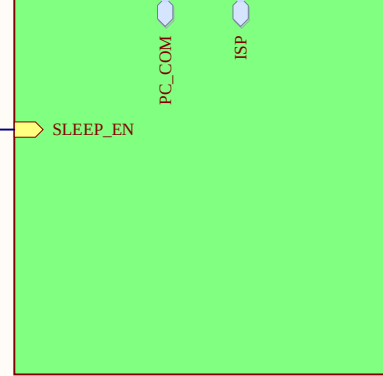
Peripherals
peep_io.SchDoc



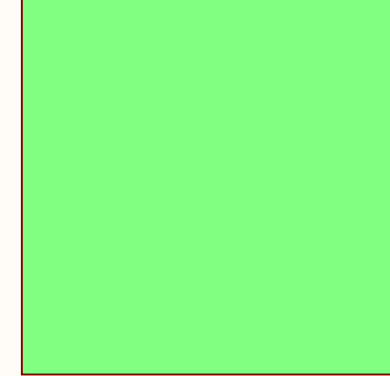
Power
peep_power.SchDoc



Xiao & ISP
Xiao_and_ISP.SchDoc

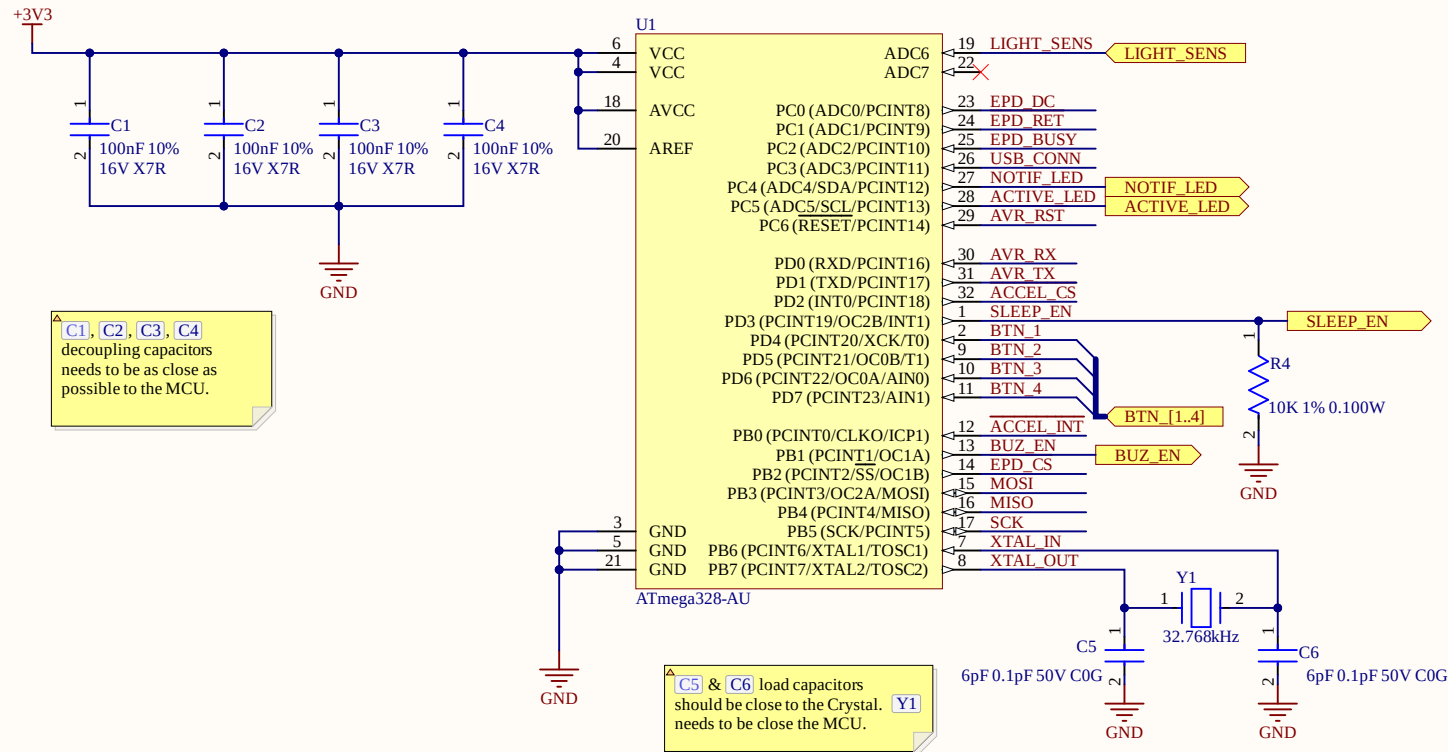


Mechanicals
Mechanicals.SchDoc

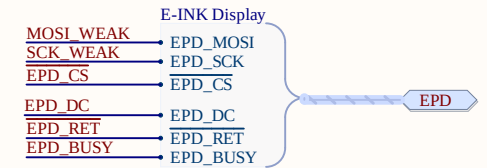
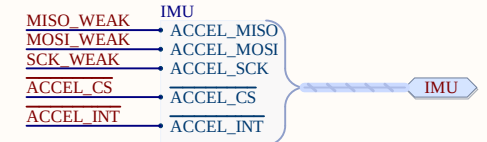


Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	1 Master	5
Date:	8/19/2025	Sheet 1 of 7
File:	Master.SchDoc	Drawn By: Arvin Dev Polayil

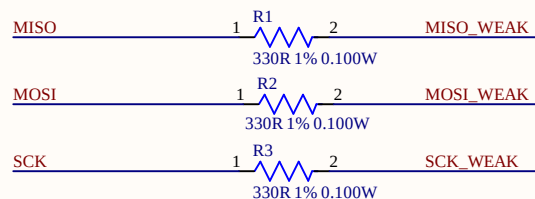
MCU



Peripheral Connections



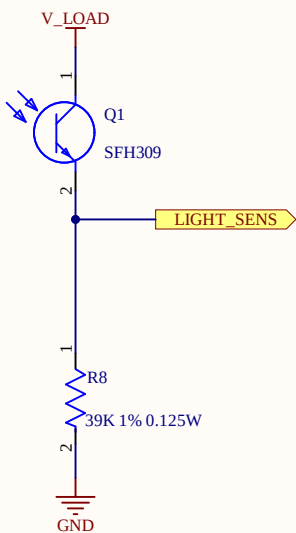
SPI



R1, R2, R3 provide 'weak' and 'strong' sides of SPI to ensure devices don't interfere during programming.

Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	2 MCU	5
Date:	8/19/2025	Sheet 2 of 7
File:	peep_avr.SchDoc	Drawn By: Arvin Dev Polayil

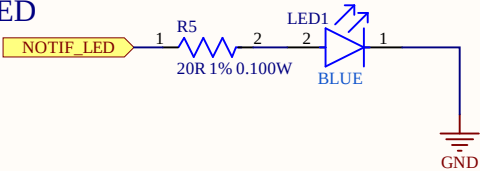
Light Sensor



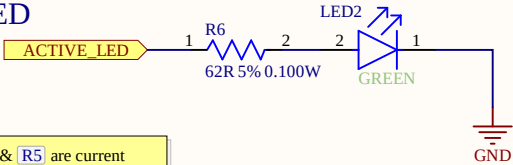
Value of R8 sets current through Q1 and adjusts the sensitivity of the light sensor.

LEDs

Notification LED



Active LED

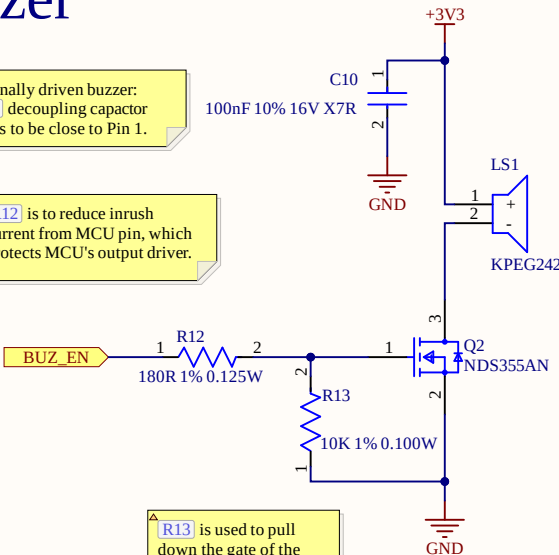


R6 & R5 are current limiting resistors for the LEDs.

Buzzer

Internally driven buzzer: C10 decoupling capacitor needs to be close to Pin 1.

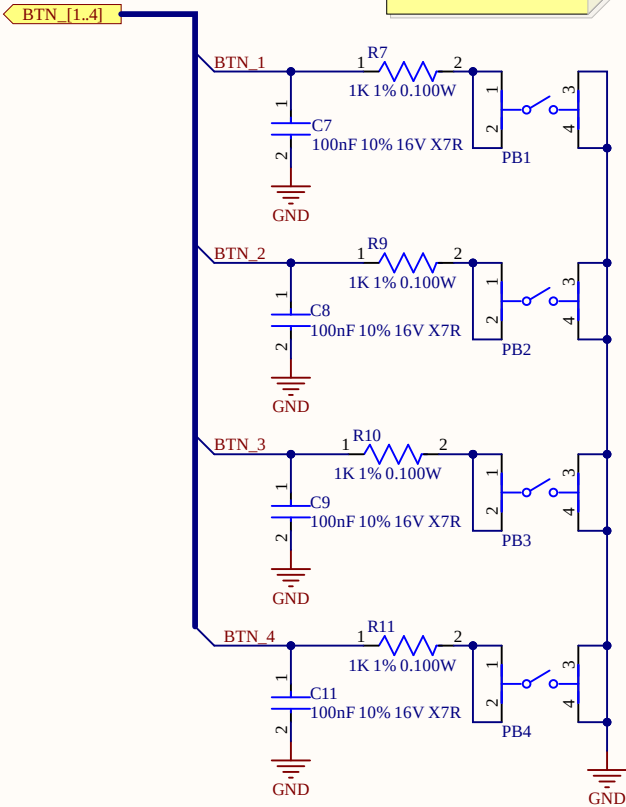
R12 is to reduce inrush current from MCU pin, which protects MCU's output driver.



R13 is used to pull down the gate of the MOSFET.

Buttons

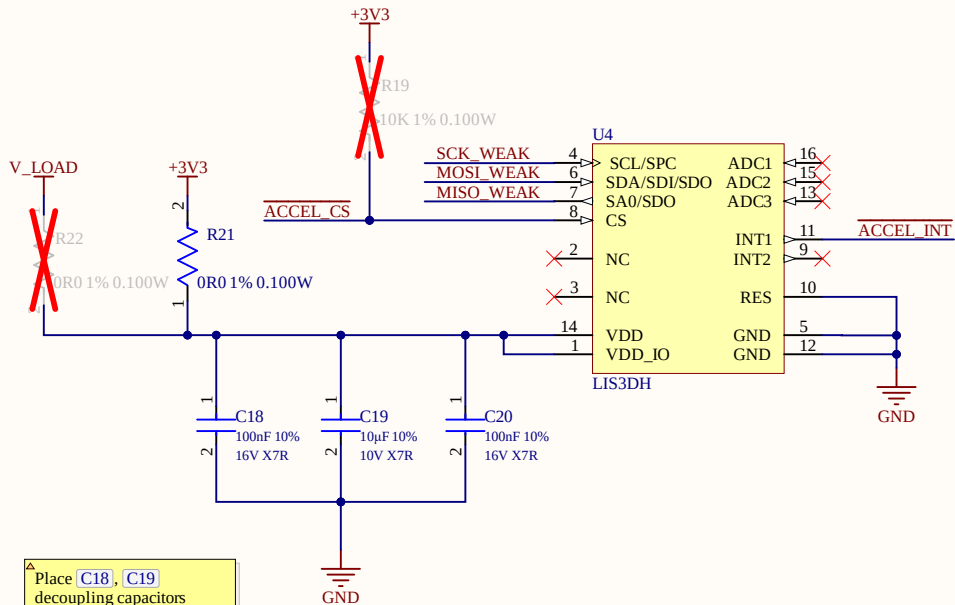
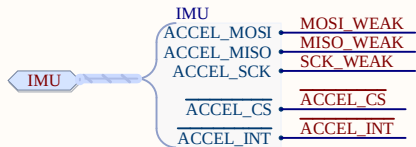
C7, C8, C9 & C11 combined with the internal pull ups of the MCU form an RC filter to debounce the buttons. Place capacitors close to button



R7, R9, R10, R11 are to slow down the discharge of the capacitor and dampen high frequency noise.

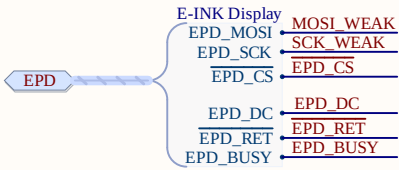
Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	3 Peep IO	5
Date:	8/19/2025	Sheet 3of 7
File:	peep_io.SchDoc	Drawn By: Arvin Dev Polayil

IMU

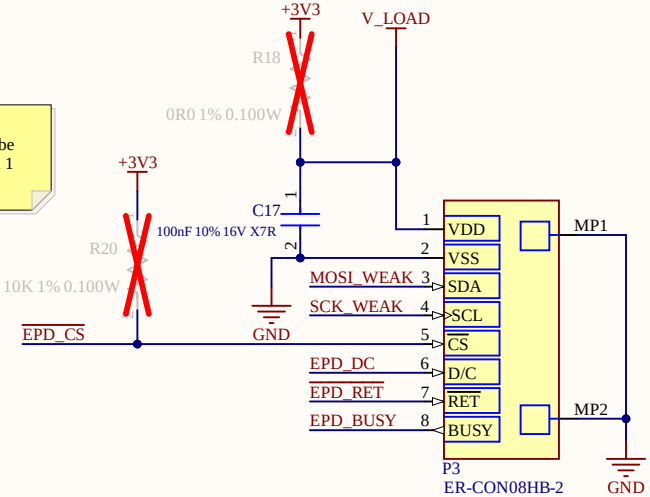


Place C18, C19 decoupling capacitors close to Pin 14, and decoupling capacitor C20 close to Pin 1.

E-Ink Display

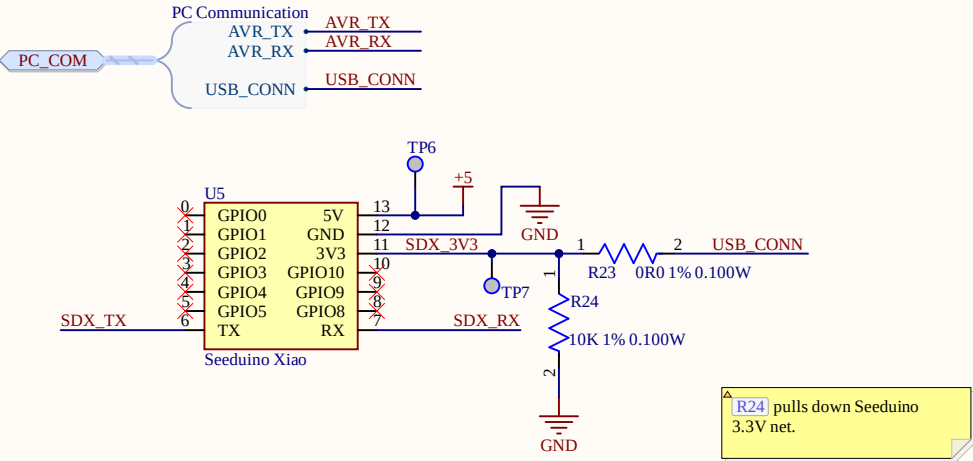


C17 decoupling capacitor needs to be placed close to Pin 1 and GND.

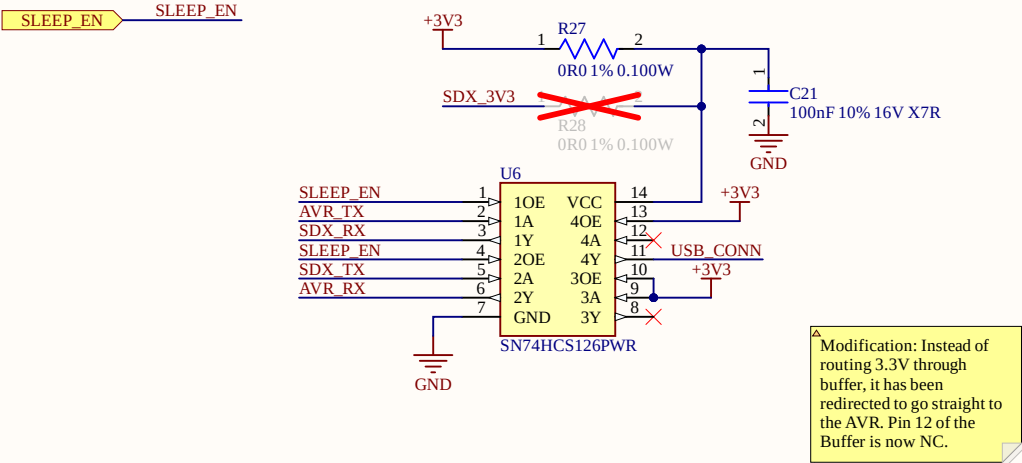


Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	4 IMU & EPD	5
Date:	8/19/2025	Sheet 4 of 7
File:	imu_and_epd.SchDoc	Drawn By: Arvin Dev Polayil

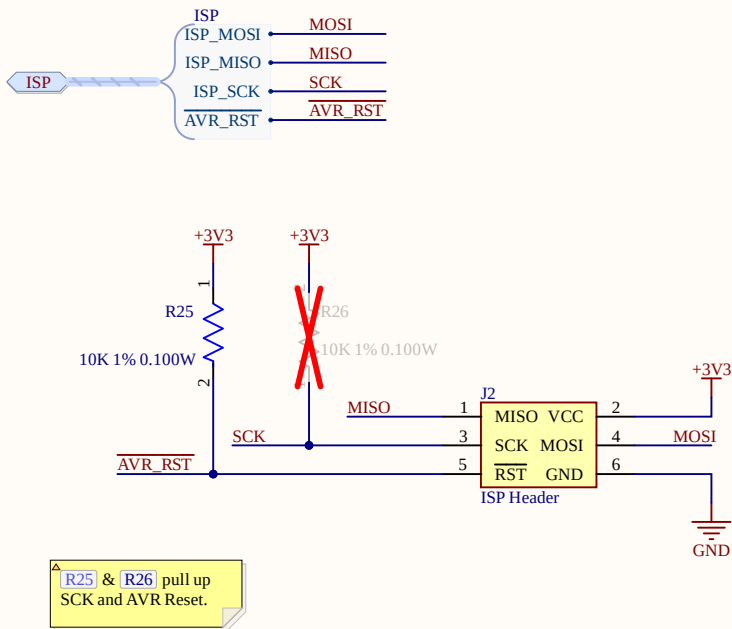
Seeduino Xiao



Seeduino Buffer

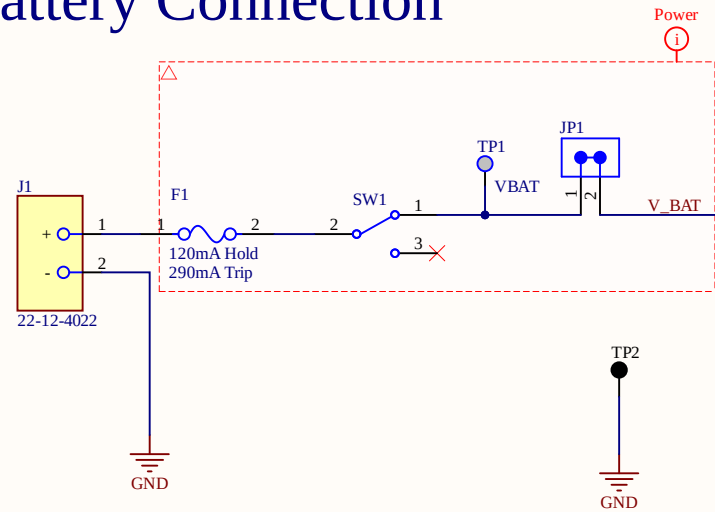


ISP Header



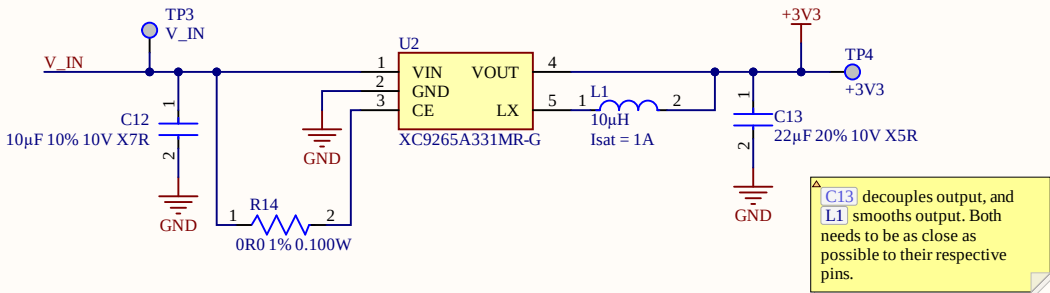
Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	5 Xiao & ISP	5
Date:	8/19/2025	Sheet5 of 7
File:	Xiao_and_ISP.SchDoc	Drawn By: Arvin Dev Polayil

Battery Connection

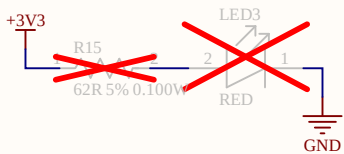


Voltage Regulation

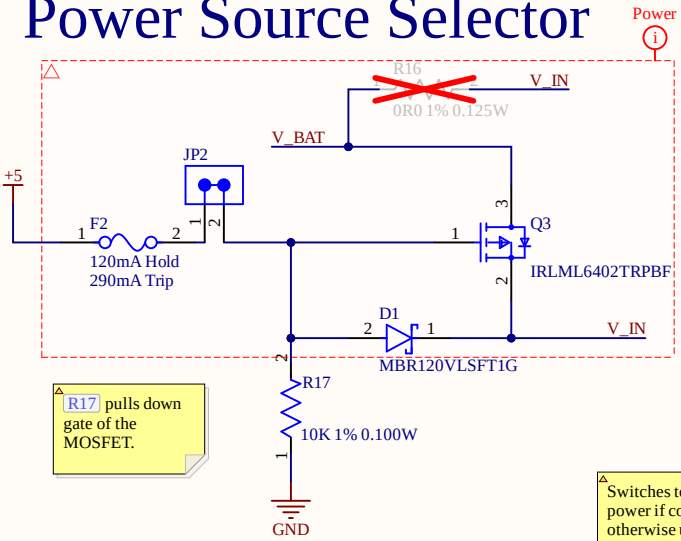
Place C12 decoupling capacitor close to Pin 1.



Power LED for debugging.

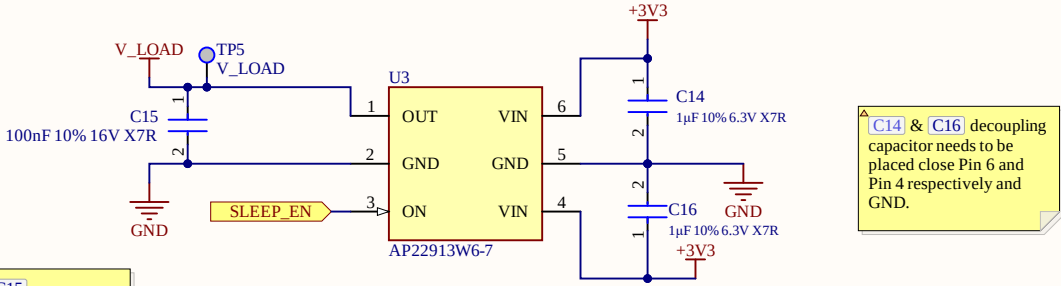


Power Source Selector



Load Switch

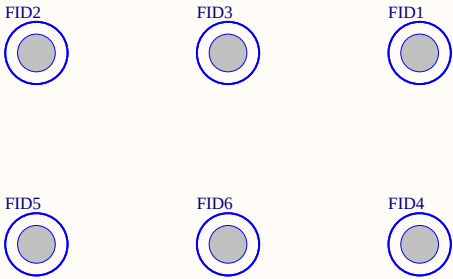
Place C15 decoupling capacitor close to Pin 1.



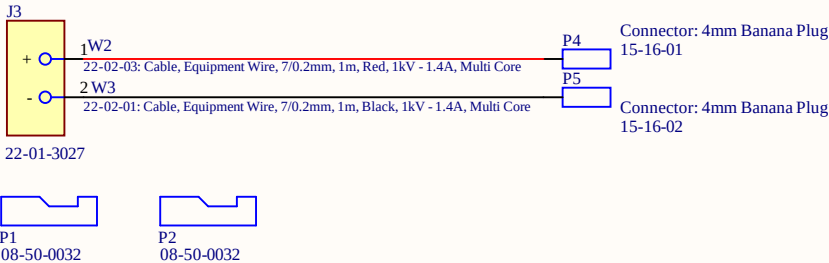
C14 & C16 decoupling capacitor needs to be placed close Pin 6 and Pin 4 respectively and GND.

Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	6 Peep Power	5
Date:	8/19/2025	Sheet 6 of 7
File:	peep_power.SchDoc	Drawn By: Arvin Dev Polayil

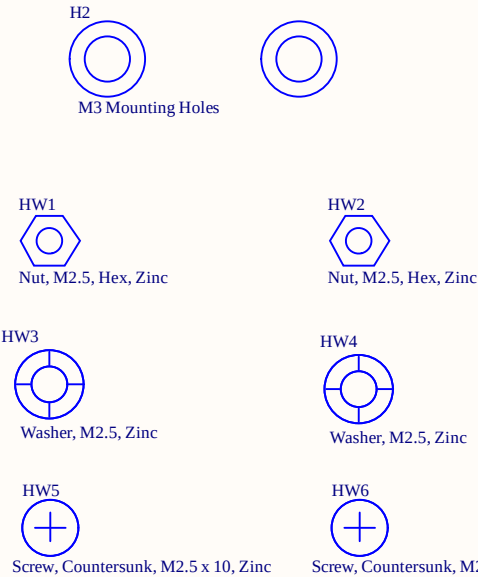
Fiducials



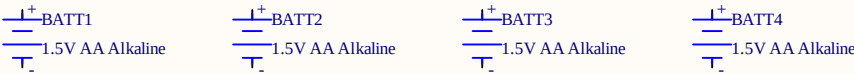
Battery Simulator Cable



Battery Mounting



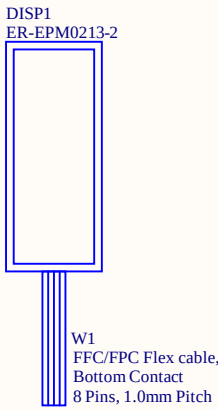
Batteries



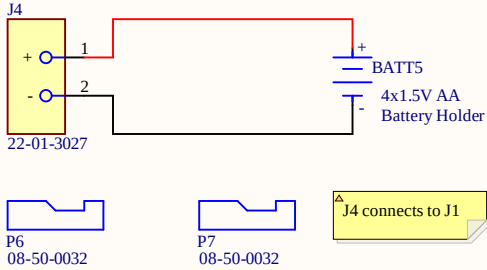
Screen Mounting



E-paper Display



Battery Holder



FFC Connects to P3

Title		
ENGG2800 Team 13: Peep		
Size	Number	Revision
A4	6 Mechanicals	5
Date:	8/19/2025	Sheet6 of 6
File:	Mechanicals.SchDoc	Drawn By: Arvin Dev Polayil